

What's in the Book

This book is divided into four parts. Part I introduces object-oriented programming:

- Chapter 1 provides a review of coding using procedural programming. I'll show you how to implement a text-based card game and simulate a bank performing operations on one or more accounts. Along the way, I discuss common problems with the procedural approach.
- Chapter 2 introduces classes and objects and shows how you can represent real-world objects like light switches or a TV remote in Python using classes. You'll see how an object-oriented approach solves the problems highlighted in the first chapter.
- Chapter 3 presents two mental models that you can use to think about what's going on behind the scenes when you create objects in Python. We'll use Python Tutor to step through code and see how objects are created.
- Chapter 4 demonstrates a standard way to handle multiple objects of the same type by introducing the concept of an object manager object. We'll expand the bank account simulation using classes, and I'll show you how to handle errors using exceptions.

Part II focuses on building GUIs with pygame:

- Chapter 5 introduces the pygame package and the event-driven model of programming. We'll build a few simple programs to get you started with placing graphics in a window and handling keyboard and mouse input, then develop a more complicated ball-bouncing program.
- Chapter 6 goes into much more detail on using OOP with pygame programs. We'll rewrite the ball-bouncing program in an OOP style and develop some simple GUI elements.
- Chapter 7 introduces the `pygwidgets` module, which contains full implementations of many standard GUI elements (buttons, checkboxes, and so on), each developed as a class.